Import Data using Transform Maps (Spreadsheet)

Phase-6 -- Project Testing Phase

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PROJECT TESTING OVERVIEW

The Testing Phase validates end-to-end behavior for: spreadsheet data loading, Import Set staging, source→target field mapping, transform execution, creation and update of employee records, duplicate prevention via Coalesce, repeated import behavior, and subsequent report and dashboard verification. Tests compare expected outcomes with actual results and document any discrepancies or errors.

- **Test Data Verification** – Verify that the spreadsheet contains the required Employee ID, Employee

Name, Email, Department, and Location fields.

- **Import Set Completion Check** – Confirm that the Import Set reaches a successful completion state after the spreadsheet is loaded.
- **Target Table Record Verification** – Check that successfully transformed records are available in the Employee Test table.
- **Field Value Verification** – Compare imported values with the original spreadsheet values to confirm that the correct information appears in the target fields.
- **New Record Testing** – Verify that newly added Employee IDs create new employee records.
- **Existing Record Testing** – Verify that existing Employee IDs are recognized and their changed Name or Email values are updated.
- **Transform History Verification** – Check the Transform History to confirm the number of uploaded, inserted, updated, and ignored records.
- **Repeated Import Verification** – Reuse the same spreadsheet and verify that unchanged records are not inserted again.
- **Report Data Verification** – Compare report results with the employee records stored in the Employee Test table.
- **Dashboard Content Verification** – Verify that the Department, Location, and Employee List reports are successfully displayed together in the Employee Analytics Dashboard.

TESTING OBJECTIVES

The testing objectives define measurable checks to confirm correct system behavior for the spreadsheet-based import workflow.

- **Verify test data consistency** – Confirm that Employee ID, Employee Name, Email, Department, and Location values are available and correctly structured.
- **Verify Import Set completion status** – Confirm that the spreadsheet import reaches a successful completion state.
- **Verify staging data accuracy** – Confirm that records loaded into the Employee Import table match the spreadsheet data.
- **Verify target field values** – Confirm that transformed records contain the correct values in the Employee Test table.
- **Verify Employee ID identification** – Confirm that Employee IDs are correctly used to identify existing employee records during testing.
- **Verify changed employee information** – Confirm that modified Name and Email values are correctly reflected for existing employees.
- **Verify transformation history results** – Confirm that uploaded, inserted, updated, and ignored record counts are correctly displayed.
- **Verify repeated-import consistency** – Confirm that importing the same spreadsheet again does not create unnecessary duplicate records.
- **Verify report accuracy** – Confirm that Department, Location, and Employee List reports represent the records in the Employee Test table.
- **Verify dashboard information** – Confirm that the required reports are successfully available in the Employee Analytics Dashboard.

TESTING ENVIRONMENT

Testing Component	Configuration	Platform	Notes
Source File	Sample Spreadsheet.xlsx	Spreadsheet	Sheet 1, Header Row = Row 1
Import Set Table	Employee Import (u_employee_import)	Import Set	Staging for incoming rows
Transform Map	Sample Spreadsheet Import	Transform	Maps u_employee_import → u_employee_test
Target Table	Employee Test (u_employee_test)	Target	Holds inserted/updated employee records
Unique Identifier	Employee ID	Coalesce	Duplicate prevention key
Dashboard	Employee Analytics	ServiceNow Dashboard	Contains three reports

These components compose the testing environment used to validate import, transformation, and reporting behaviors.

TEST DATA PREPARATION

The test spreadsheet includes columns: Employee ID, Employee Name, Email, Department, and Location. Test data is prepared to exercise inserts and updates:

- **Define required columns** – Ensure the test spreadsheet contains Employee ID, Employee Name, Email, Department, and Location.
- **Prepare unique Employee IDs** – Include new Employee IDs to specifically test record insertion.
- **Prepare existing Employee IDs** – Include Employee IDs that already exist in the Employee Test table.
- **Modify existing employee details** – Change Name or Email values for existing Employee IDs to test update behavior.
- **Include department information** – Provide Department values so the imported records can be validated against the department report.
- **Include location information** – Provide Location values for testing the location-based report.
- **Maintain spreadsheet structure** – Keep the required column structure consistent during repeated testing.
- **Prepare four-row test data** – Use the documented scenario containing two new records and two existing records with changed information.
- **Retain the original test spreadsheet** – Keep the same spreadsheet available for the repeated- import test.
- **Prepare data for repeated-import validation** – Reuse the same four-row spreadsheet after the first transformation to verify that the records are handled consistently and that duplicate records are not created.

Documented test scenario contains four rows: two new employee records and two existing employee records with changed Name or Email values.

Test Data Flow

Test Spreadsheet

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2 New Employee IDs + 2 Existing Employee IDs

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Employee Import Staging Table

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Transform Map

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2 Inserted + 2 Updated

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Repeat Same Spreadsheet

↓

4 Ignored

IMPORT SET TESTING

First activity: verify spreadsheet data loads into the Employee Import staging table and that the Import Set completes successfully.

Test ID	Test Case	Expected Result	Actua l Resul t	Status
IST-01	Load Sample Spreadsheet.xlsx	File loads successfully	File loade d succe ssfull y	Pass
IST-02	Select Sheet 1	Correct sheet is selected	Sheet 1 select ed	Pass
IST-03	Set Header Row to 1	Spreadsheet columns recognized	Colu mns recog nized	Pass
IST-04	Load data into Employee Import	Records appear in staging table	Recor ds loade d	Pass
IST-05	Check Import Set state	Import Set completes successfully	Comp leted and succe ssful	Pass

TARGET TABLE VALIDATION

Verify Employee Test target table both before and after transformation. Confirm required fields, presence of imported values, and record visibility.

Test ID	Validation	Expected Result	Status
TT-01	Employee ID field	Employee ID displayed correctly	Pass
TT-02	Employee Name field	Employee Name displayed correctly	Pass
TT-03	Email field	Email displayed correctly	Pass
TT-04	Department field	Department displayed correctly	Pass
TT-05	Location field	Location displayed correctly	Pass
TT-06	Imported records	Records appear in Employee Test	Pass
TT-07	Updated records	Changed values are reflected	Pass
TT-08	New records	New Employee IDs appear	Pass
TT-09	Data comparison	Target values match spreadsheet data	Pass
TT-10	Record visibility		Pass

		Imported records are clearly visible	

TRANSFORM MAP TESTING

Testing of the "Sample Spreadsheet Import" Transform Map focuses on correct source/target selection, field mapping, successful execution, and expected record creation or update.

Test ID	Test Case	Expected Result	Actual Result	Status
TM-01	Verify source table Employee Import	Correct source selected	Correct source selected	Pass
TM-02	Verify target table Employee Test	Correct target selected	Correct target selected	Pass
TM-03	Verify field mapping	Fields map correctly	Fields mapped correctly	Pass
TM-04	Run Transform	Transform completes successfully	Transform completed	Pass
TM-05	Check Employee Test		Records available	Pass

		Records appear in target table		

COALESCE, INSERT & UPDATE TESTING

Coalesce logic is tested to ensure Employee ID identifies existing records and prevents duplicates. The main test spreadsheet contains four rows: two new Employee IDs and two existing Employee IDs with modified Name/Email. Expected outcome: two inserts and two updates.

Test ID	Test Scenario	Expected Result	Actual Result	Status
CO-01	Import new Employee ID	New record inserted	New record inserte	Pass
CO-02	Import existing Employee ID	Existing record identified	Existing record identified	Pass
CO-03	Existing ID with changed Name	Existing record updated	Record updated	Pass
CO-04	Existing ID with changed Email	Existing record updated	Record updated	Pass
CO-05	Repeat same spreadsheet	No duplicate records created	Records ignored	Pass

Main Insert / Update Results

Test Scenario	Input	Expected Result	Status
New Employee 1	New Employee ID	Insert	Pass

New Employee 2	New Employee ID	Insert	Pass
Existing Employee 1	Existing ID + changed data	Update	Pass
Existing Employee 2	Existing ID + changed data	Update	Pass

Total 4 rows were uploaded. 2 rows were inserted and 2 rows were updated.

REPEATED IMPORT, HISTORY, DATA, REPORT & DASHBOARD VALIDATION

Repeated import testing, transform history checks, data validation, report verification, and dashboard validation confirm overall integrity and system behavior for repeated submissions and reporting.

Repeated Import Testing

Sure — here is the clean **table format** you can directly copy into Word, Excel, or your test report.

Repeated Import Testing

Test ID	Test Scenario	Expected Result	Actual Result	Status
RI-01	Import same spreadsheet again	Existing records recognized	Existing records recognized	Pass
RI-02	Check inserted count	0 new records	0 inserted	Pass
RI-03	Check updated count	0 unchanged records updated	0 updated	Pass
RI-04	Check ignored count	Duplicate/unchanged rows ignored	4 ignored	Pass

Transform History Validation

Test ID	Transform History Check	Expected Result	Actual Result	Status
TH-01	Check Import Employee Records	No duplicate records	Completed successfully — no duplicate records	Pass
TH-02	Uploaded rows	Correct number displayed	Correct number displayed	Pass
TH-03	Inserted rows	Matches test data	Matches test data	Pass
TH-04	Updated rows	Matches existing records	Matches existing records	Pass

TH-05	Ignored rows	Correct for repeated import	Correct for repeated import	Pass

Summary

Test Area	Uploaded Rows	Inserted	Updated	Ignored	Status
Repeated Import	4	0	0	4	Pass
Transform History	4	0	0	4	Pass

Data Validation Testing

Test ID	Data Validation	Expected Result / Status
DV-01	Employee ID	Correct ID displayed — Pass
DV-02	Employee Name	Correct name displayed — Pass
DV-03	Email	Correct email displayed — Pass
DV-04	Department	Correct department displayed — Pass
DV-05	Location	Correct location displayed — Pass
DV-06	Updated records	Modified values displayed — Pass
DV-07	New records	New employees displayed — Pass
DV-08	Duplicate check	No duplicate records — Pass

Report Testing

Three reports were validated: Employees by Department, Employees by Location, and Employee List Report. Tests confirm correct source, grouping, aggregation, and fields.

Test ID	Report	Expected Result / Status
RP-01	Employees by Department	Department-wise employee count displayed — Pass

RP-02	Employees by Location	Location-wise employee count displayed — Pass
RP-03	Employee List Report	Employee details displayed correctly — Pass
RP-04	Report data	Reflects Employee Test data — Pass

Dashboard Testing

Test ID	**Dashboard Test**	**Expected Result / Status**
DB-01	Open Employee Analytics Dashboards	Dashboard opens successfully — Pass
DB-02	Check Department report	Report is displayed — Pass
DB-03	Check Location report	Report is displayed — Pass
DB-04	Check Employee List report	Report is displayed — Pass
DB-05	Verify dashboard data	Data matches Employee Test table — Pass

Complete Testing Summary

Testing Area	Test Cases	Result
Import Set Testing	5	Passed
Target Table Validation	5	Passed
Transform Map Testing	5	Passed
Coalesce Testing	5	Passed
Insert/Update Testing	4	Passed
Repeated Import Testing	5	Passed
Transform History Testing	5	Passed
Data Validation	8	Passed
Report Testing	4	Passed
Dashboard Testing	5	Passed

The table above summarizes the principal testing areas executed during the Testing Phase.

Testing Results

- Spreadsheet imported successfully.
- Import Set completed successfully.
- Transform Map executed successfully.
- Employee records were created in Employee Test.
- Two new employee records were inserted; two existing records were updated.
- Coalesce prevented duplicate insertion.
- Repeated import resulted in 4 ignored records.
- Employee Test table values validated.
- Three reports were created and tested and added to the Employee Analytics Dashboard.